Tainã Coleman

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Education

University of Southern California — Los Angeles, CA
Ph.D. in Computer Science; GPA: 4.0 — *2020 - 2023*
Dissertation: Scientific Workflows: Generation and Benchmarking

California State University, Long Beach — Long Beach, CA
Master of Science in Computer Science; GPA: 4.0 — *2018 - 2020*
Thesis: SharkID: A Framework for Automated Individual Shark Identification

Universidade Federal de Itajubá — Itajubá, Minas Gerais, Brazil
BS Computer Engineering — *2011 - 2016*

Experience

Loyola Marymount Unviversity — Los Angeles, CA
Assistant Professor in Computer Science — *Aug 2026 - Present*

San Diego Supercomputer Center — San Diego, CA
Schmidt AI in Science Postdoctoral Fellow — *Jan 2024 – Aug 2026*

- Lead a team advancing scientific AI workflows, focusing on performance, accuracy, and reproducibility across large-scale data and compute systems.
- Contribute to benchmarking efforts in collaboration with the WIFIRE Lab and Schmidt AI in Science Fellows.
- Support grant writing to secure funding for research on scalable and reproducible AI in science.

Lawrence Livermore National Laboratory — Livermore, CA
Graduate Student Research Assistant — *Jan 2023 – Dec 2023*

- Designed high-performance workflow components enabling GPU memory transfers and GPU Direct communication.
- Developed primitives for modeling storage- and network-based communication patterns in scientific workflows.

The Aerospace Corporation — El Segundo, CA
Embedded Control Systems Graduate Intern — *May 2022 – Dec 2023*

- Built and optimized pre-launch certification workflows for multiple launch programs.
- Developed visualization and management tools to streamline workflow execution.
- Automated legacy simulation components to improve reliability and reduce manual overhead.

Awards & Recognition

Rising Stars in Computational and Data Sciences
2023 Rising Stars in Computational and Data Sciences Workshop — *2023*

Best Graduate Research Assistant Award
Viterbi Ph.D. Awards Ceremony — *2022*

Works in Progress / Under Review

WfCluster: Using WFChef Patterns to Improve Workflow Scheduling
T. Coleman, *J. Coleman, R. Ferreira da Silva, F. Suter*
In Preparation

Automated Task Partitioning of Workflows for Hybrid HPC-Serverless Platforms
V. Thurimella, A.A.Da Silva, G. Rattihalli, ***T. Coleman****, R. Ferreira Da Silva, D. Milojicic*
In Preparation

Publications

[18] **ARGUS: Agentic Researcher Guide for Unified Science From Natural Language Questions to Reproducible Scientific Results**
K. Lin, M. Nguyen, ***T. Coleman****, I. Altintaş.*
To apperar at the IEEE International Conference on eScience, 2026

[17] **Agents of Intelligence in Practice: A Multi-Agent Model Recommendation System for the National Data Platform**
T. Coleman*, H. Ahmed, I.Gosh, I. Perez, K. Lin, M. Nguyen, T. Javidi, I. Altintaş.*
To apperar at the IEEE International Conference on eScience, 2026

[16] **Serverless Workflow Benchmarking with HSMFlow and WfBench**
A.A. Da Silva, ***T. Coleman****, F. Sutter, R. Ferreira Da Silva, G. Rattihalli, and V. Thurimella.*
26th ACM/IFIP International Middleware Conference

[15] **Toward Agents of Intelligence: Bridging the AI Expertise Gap in Domain Sciences**
T. Coleman*, I. Altintaş*
5th Workshop on Reproducible Workflows, Data Management, and Security in conjunction with e-Science25

[14] **A Terminology for Scientific Workflow Systems**
F. Suter, ***T. Coleman****, I. Altintaş, ..., R. Ferreira da Silva*
Future Generation Computer Systems (FGCS), 2025. DOI: 10.1016/j.future.2025.107974

[13] **WfCommons - a framework for enabling scientific workflow research and development (Tutorial)**
R. Ferreira da Silva, ***T. Coleman****, F. Suter, H. Casanova*
Humboldt University of Berlin, 2025

[12] **BanditWare: A Contextual Bandit-based Framework for Hardware Prediction**
T. Coleman*, H. Ahmed, R. Shende, I. Perez, I. Altintaş*
AI4Sys @ Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2025

[11] **WfCommons - a framework for enabling scientific workflow research and development (Tutorial)**
R. Ferreira da Silva, ***T. Coleman****, F. Suter, H. Casanova*
IEEE International Conference on eScience, 2024

[10] **Automated Generation of Scientific Workflow Generators with WfChef**
T. Coleman*, H. Casanova, R. Ferreira da Silva*
Future Generation Computer Systems (FGCS), 2023. DOI: 10.1016/j.future.2023.04.031

[9] **WfBench: Automated Generation of Scientific Workflow Benchmarks**
T. Coleman*, H. Casanova, K. Maheshwari, L. Pottier, S. R. Wilkinson, J. Wozniak, F. Suter, M. Shankar, R. Ferreira da Silva*
IEEE/ACM PMBS, 2022. DOI: 10.1109/PMBS56514.2022.00014

[8] **WfCommons: A Framework for Enabling Scientific Workflow Research and Development**
T. Coleman*, H. Casanova, L. Pottier, M. Kaushik, E. Deelman, R. Ferreira da Silva*
Future Generation Computer Systems, 2022. DOI: 10.1016/j.future.2021.09.043

[7] **WfChef: Automated Generation of Accurate Scientific Workflow Generators**
T. Coleman*, H. Casanova, R. Ferreira da Silva*
IEEE International Conference on eScience, 2021

[6] **A Community Roadmap for Scientific Workflows Research and Development**
R. Ferreira da Silva, H. Casanova, K. Chard, I. Altintaş, R. M. Badia, B. Balis, ***T. Coleman****, ..., M. Wolf*
IEEE WORKS, 2021

[5] **Evaluating Energy-Aware Scheduling Algorithms for I/O-Intensive Scientific Workflows**
T. Coleman*, H. Casanova, T. Gwartney, R. Ferreira da Silva*
International Conference on Computational Science, 2021. DOI: 10.1007/978-3-030-77961-0_16

[4] **Workflows Community Summit: Bringing the Scientific Workflows Community Together**
R. Ferreira da Silva, H. Casanova, K. Chard, ..., ***T. Coleman****, ..., J. Wozniak*
arXiv:2103.09181, 2021

[3] **Workflows Community Summit: Advancing the State-of-the-art of Scientific Workflows Management Systems Research and Development**
*R. Ferreira da Silva, H. Casanova, K. Chard, **T. Coleman**, D. Laney, D. Ahn, ..., J. Wozniak*
arXiv:2106.05177, 2021

[2] **WorkflowHub: Community Framework for Enabling Scientific Workflow Research and Development**
*R. Ferreira da Silva, L. Pottier, **T. Coleman**, E. Deelman, H. Casanova*
IEEE WORKS, 2020. DOI: 10.1109/WORKS51914.2020.00012

[1] **A Biometric for Shark Dorsal Fins Based on Boundary Descriptor Matching**
***T. Coleman**, J. Moon*
CAINE (International Conference on Computer Applications in Industry and Engineering), 2019

Projects

WfCommons (wfcommons.org)
An open-source toolkit for analyzing, synthesizing, and benchmarking scientific workflows *2020–Present*

National Data Platform (nationaldataplatform.org)
A federated and extensible data ecosystem to promote collaboration, innovation, and use of data on top of existing cyberinfrastructure capabilities. *2024–Present*